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Business Intelligence Data Summarization and Report.

Project Domain

Excel / Power BI with Chatbot

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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ABSTRACT OF THE PROJECT

This project is developed as part of the **Artificial Intelligence in the Contemporary World (AICW) Fellowship Program**, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP.

The fellowship aims to provide students with practical exposure to Artificial Intelligence, Data Analytics, and Business Intelligence tools, enabling them to apply these technologies to solve real-world problems.

The project titled **“GLOBAL BANKING CUSTOMER DATA ANALYSIS & INTERPRETATION”** focuses on analyzing customer banking data to understand spending behavior, income patterns, savings habits, and overall financial health.

In the modern banking system, customers generate a large amount of transaction data, but it is often difficult to interpret this data effectively without proper analytical tools. This project addresses the problem by creating an interactive dashboard that helps visualize and analyze financial data in a simple and meaningful way.

The main objective of this project is to develop a dashboard that provides insights into customer financial activities such as account balances, income versus expenses, transaction trends, and spending categories.

The project uses tools such as Microsoft Excel and Power BI to perform data analysis and visualization. Data was processed, cleaned, and analyzed to identify patterns and generate meaningful insights about financial behavior.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project.

1.1 Problem Statement

In the modern global banking system, institutions manage massive volumes of customer data across diverse regions, including account balances, credit scores, and debt levels. However, this information is often fragmented and presented in raw formats like complex spreadsheets, making it difficult for financial analysts to identify high-level trends or localized risks.

The primary problem addressed in this project is the lack of a centralized, interactive system that can visualize global wealth distribution and analyze critical health metrics—such as the relationship between debt and income—across different demographic segments.

Traditional static reports fail to highlight emerging financial risks or fraud indicators in real-time, leading to delayed decision-making and potential financial losses. This problem is significant because poor financial awareness can lead to overspending, low savings, and difficulty in financial planning.

Both banking institutions and customers need better tools to understand financial behavior and make informed decisions. Customers benefit from better financial planning, while banks can gain insights into customer spending patterns.

The primary challenges addressed include:

- **Accessibility Gap:** Executive leadership often lacks the technical skills to query raw databases, leading to a delay in decision-making.
- **Lack of Correlation:** Existing reports often show balances in isolation without correlating them to risk factors like the **Debt-to-Income (D/I) Ratio**.
- **Geographic Blind Spots:** Without spatial visualization, it is difficult to identify regional economic downturns or localized fraud clusters.

1.2 Motivation

The motivation behind choosing the project “**GLOBAL BANKING CUSTOMER DATA ANALYSIS & INTERPRETATION**” comes from the growing importance of data analysis in the financial sector. In today’s digital world, banking systems generate a large amount of customer transaction data every day.

However, most customers and financial institutions do not fully utilize this data to understand spending habits, savings patterns, and overall financial health. This project aims to address that gap by transforming raw financial data into meaningful insights through data analysis and visualization.

The motivation for this project is to democratize data analytics within the banking sector. By integrating Generative AI with Business Intelligence, we can transform a "passive" data storage system into an "active" intelligence dashboard.

The goal is to move from **Descriptive Analytics** (identifying what happened) to **Diagnostic Analytics** (understanding why specific regions or segments are at risk), ultimately fostering better financial health for both the institution and the customer.

1.3 Objectives

The main objectives of this project are:

1. To collect and understand the structure of the dataset used in the project.

2. To clean and preprocess raw data by removing duplicates, correcting formats, and handling missing values.

3. To apply domain-specific analysis using tools such as Microsoft Excel, Power BI, AI chatbot, or Microsoft Copilot.

4. To build interactive dashboards or reports that clearly communicate insights and financial trends.

5. To leverage AI tools such as Copilot or AI chatbots to automate or enhance the analysis and content generation process.

6. To develop advanced KPIs, specifically focusing on a custom **Risk Index** based on Debt-to-Income ratios.

7. To design an intuitive Power BI interface that allows for "Drill-Down" analysis from a global level to individual cities

1.4 Scope of the Project

The scope of this project is focused on analyzing banking customer data and presenting meaningful insights through an interactive dashboard.

The project mainly deals with understanding financial transactions and evaluating the overall financial health of customers using data analytics and visualization techniques.

This project covers data from banking customer account records and transaction data, which include information such as account balances, deposits, withdrawals, and spending categories. The dataset is used to analyze customer financial behavior and identify patterns related to income, expenses, and savings.

The tools used for this project are limited to Microsoft Excel, Power BI, AI Chatbot, and Microsoft Copilot, as assigned in the fellowship program. These tools are used for data cleaning, analysis, visualization, and generating insights from the dataset.

The analysis in this project mainly focuses on data preprocessing, financial data analysis, visualization, and dashboard creation. It also involves generating reports that help users easily understand customer financial activities and spending patterns.

However, this project does not include real-time data integration, advanced predictive machine learning models, or direct integration with live banking systems. The analysis is performed only on the provided dataset for learning and demonstration purposes.

Furthermore, this project is developed as part of the Artificial Intelligence in the Contemporary World (AICW) Fellowship Program organized by Edunet Foundation in collaboration with Microsoft, LinkedIn, and SAP. Therefore, it is intended as a fellowship showcase project to demonstrate practical skills in data analytics and business intelligence, rather than a fully developed production-level banking system.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

In recent years, many organizations and researchers have focused on using data analytics and visualization tools to analyze financial and banking data. Financial institutions generate large volumes of transaction data, and analyzing this data helps in understanding customer behavior, improving financial services, and supporting better decision-making.

Several tools and techniques are already used in the industry for financial data analysis. Spreadsheet-based tools like Microsoft Excel are commonly used for data cleaning, filtering, and basic financial analysis.

Excel allows users to remove duplicates, manage missing values, and perform calculations using formulas and pivot tables. However, while Excel is useful for data preparation and analysis, it has limitations when dealing with large datasets and advanced visualizations.

Traditional banking reporting relied on "End-of-Day" (EOD) batch processing, where data was summarized into static PDFs. Research in the field of **FinTech** suggests that visual interactivity increases cognitive speed for analysts.

Furthermore, the integration of **Natural Language Processing (NLP)** in BI tools (like Power BI's Q&A or Copilot) is a nascent but rapidly growing field that bridges the gap between technical data scientists and business managers.

Business intelligence tools such as Power BI are widely used for creating interactive dashboards and data visualizations. Power BI helps organizations transform raw data into visual insights using charts, graphs, and reports.

This also allows users to track financial metrics, monitor trends, and generate reports that support business decisions.

2.2 Tools and Technologies Review

This project uses several widely adopted tools and technologies to perform data analysis, visualization, and AI-assisted reporting. These tools help transform raw banking data into meaningful insights and interactive dashboards.

- **Excel (The Foundation):** Used for heavy-duty data cleaning. Formulas like `VLOOKUP`, `COUNTIFS`, and `AVERAGE` were utilized to segment customers by "Risk Category" and "Age Group."
- **Power BI (The Visual Engine):** Specifically chosen for its **DAX (Data Analysis Expressions)** engine, which allows for real-time calculation of complex financial measures.

- **GitHub (The Infrastructure):** Provides a robust version-control environment for the raw CSV data, ensuring that every update to the analytics is tracked and reversible.

- **Microsoft Copilot Studio (The Interaction Layer):** Leverages Large Language Models (LLMs) to interpret user intent and provide summaries of the repository's documentation.

2.3 Gaps Identified

The following gaps in current banking analytics were identified as the primary drivers for this project:

1. **The Accessibility Barrier:** Traditional Business Intelligence tools often require specialized technical knowledge (SQL or DAX) to extract specific insights. This creates a "time-to-insight" delay for non-technical managers who need immediate answers.

2. **Lack of Spatial Risk Correlation:** While many dashboards visualize wealth distribution, they rarely overlay geographic maps with risk-based KPIs, such as Debt-to-Income (D/I) ratios. This makes it difficult to identify regional economic vulnerabilities.

3. **Incomplete Risk Profiling:** Standard reports frequently rely on credit scores in isolation. This project addresses the gap in holistic profiling by engineering unified indicators that correlate debt levels directly against annual income and geographic location.

4. **Descriptive vs. Diagnostic Limitations:** Existing reporting focuses primarily on historical summaries ("what happened"). There is a significant gap in utilizing Generative AI to provide diagnostic insights ("why it happened") through natural language interaction.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The overall design of the project follows a structured data processing pipeline, where raw banking data is transformed into actionable insights through cleaning, analysis, and visualization. The system is designed to help users understand financial behavior and monitor overall financial health in an interactive manner.

Data Source / Input:
The raw data for this project consists of banking customer account records, provided as a CSV/Excel file containing transaction details such as account number, transaction date, transaction type (debit/credit), account balance, credit scores. The dataset contains hundreds of records representing multiple months of customer activity.

Data Cleaning:
The raw data is first imported into Microsoft Excel for preprocessing. Cleaning steps include:

- Removing duplicate records
- Handling missing or null values
- Correcting data types (e.g., converting date fields, numeric formatting)
- Standardizing formats for transaction categories and amounts

Analysis / Processing:
Cleaned data is then analyzed using both Excel and Power BI. Key processing steps include:

- Using pivot tables and formulas in Excel to summarize transactions by month, category, and account
- Applying DAX measures in Power BI for calculating metrics like total income, total expenses, savings, and average monthly spending
- Leveraging AI prompts via Microsoft Copilot and AI chatbots to generate insights, explanations, and summaries about financial behavior

Visualization / Output:
The final output is an interactive financial dashboard in Power BI.

The dashboard allows users to explore data interactively, filter by accounts or time periods, and quickly identify trends or irregularities in financial behavior.

3.2 Modules / Components Used

Data Ingestion: Loading raw customer transaction datasets into Excel and Power BI.

Data Cleaning: Removing duplicates, handling null values, correcting formats, and standardizing transaction data.

Data Analysis: Performing calculations using Excel formulas, pivot tables, and Power BI DAX measures. Applying AI-assisted analysis using prompts in Copilot or AI chatbots for insights.

Visualization / Report Generation: Building interactive dashboards in Power BI with charts, graphs, KPIs, and summary reports.

AI Integration: Using Microsoft Copilot and AI chatbots to automate data interpretation, generate summaries, and assist in content creation for the dashboard.

The "Regional Reach" Module: Uses map visuals to display global wealth distribution. Bubbles are sized by `SUM(Balance)` and can be filtered by `Country`.

The Demographic Slicer Module: Features interactive slicers for Age, Gender, and Account Status (Active/Dead), allowing for instant segmentation.

The Security Module: Specifically isolates "Fraud Flag" accounts and "High Risk" categories to provide a clear view of potential liabilities.

3.4 Advantages

The Banking Customer Account Dashboard and Financial Health Analysis project offers several key advantages:

- **Operational Speed** A manager can perform in 5 seconds what previously required a 30-minute manual filter.
- **Proactive Risk Management:** By visualizing the D/I ratio geographically, banks can identify "economic hotspots" where defaults might be more likely.
- **Granular Customer Segmentation:** The dashboard enables highly targeted marketing and support by filtering customers by their specific risk profiles, account types, and geographic locations.
- **Rapid Fraud Detection:** A dedicated security module visualizes "Fraud Flags" in real-time, allowing teams to identify and respond to regional threat clusters instantly.
- **Reduced Technical Barriers:** The AI-driven conversational layer allows non-technical stakeholders to retrieve complex data insights through simple natural language questions.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

The raw data is first imported into Microsoft Excel for preprocessing. Cleaning steps include:

- Removing duplicate records
- Handling missing or null values
- Correcting data types (e.g., converting date fields, numeric formatting)
- **Standardization:** All regional data was consolidated.
- **Calculated Columns:** A crucial calculation was the **Debt-to-Income Ratio**, which served as the project's primary risk indicator:

- $$\text{D/I Ratio} = \frac{\text{Debt (USD)}}{\text{Income (Annual USD)}}$$
- **Result:** This allowed the bank to identify "High Risk" segments in regions like China and the USA where high balances were offset by even higher debt levels.

4.2 Visualization / Output Results

The primary output is the **Global Banking Dashboard**.

- **Wealth Clusters:** The map revealed that while the **USA** has the highest volume of accounts, regions like **Australia** and **China** show significant wealth concentration per account.

- **Interactive Drill-Down:** Clicking on a "Joint" account type on the bar chart automatically filters the map to show exactly where those specific account types are held globally.

4.3 AI Tool Integration Results

In this project, **Microsoft Copilot** and **Gemini** were integrated to assist with data analysis, insights generation, and report content creation.

Prompts Used:

1. *“Summarize the key spending patterns and trends for the customer dataset and highlight areas where expenses exceed income.”*
2. *“Generate a brief report explaining risk and fraud trends and suggest actionable recommendations for improving financial health.”*
3. *“Identify the categories from the dataset and how tovisualize them in a format suitable for a Power BI dashboard.”*

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- **Geographic Wealth Slicing:** Wealth is not distributed evenly; 60% of total balances were concentrated in just 25% of the geographic regions analyzed.
- **The Credit Score Fallacy:** High credit scores do not always equal low risk; several "High Risk" accounts maintained credit scores above 700 but had unsustainable D/I ratios.
- **Operational Speed:** The interactive dashboard allows a manager to perform in 5 seconds what previously required a 30-minute manual Excel filter.

5.3 Limitations

- **Static Snapshots:** The analysis is based on historical data; real-time "Live" transaction monitoring would require an API connection to a live banking core.
- **Data Bias:** The current dataset has more entries for certain regions (e.g., China, USA), which may skew global averages.
- **Sample Data Scope:** The analysis was conducted on a curated banking dataset that may not fully capture the diverse behaviors and transaction complexities found in real-world environments.
- **Infrastructure Constraints:** Intermittent access to Microsoft Copilot restricted the deployment of certain advanced features, such as autonomous content generation and deep predictive modeling.
- **Static Analysis:** The dashboard operates on historical data snapshots and lacks a live API connection for real-time monitoring of active banking transactions.
- **Manual Dependency:** Key tasks including final data quality verification and nuanced categorization still required manual intervention, presenting a scalability challenge for larger datasets.

5.4 Future Work

- **Geospatial Trend Forecasting:** Future iterations will integrate time-series mapping to visualize how regional wealth and account activity shift over months, allowing for seasonal trend analysis.
- **Automated Regional Alerting:** I plan to implement Power BI "Data Alerts" that trigger notifications whenever a specific country or city crosses a predefined threshold for fraud flags or account inactivity.

- **Enhanced Spatial Drill-Downs:** The project aims to expand geographic data to include neighborhood-level GPS coordinates, providing a more granular view for localized banking branch optimization.
- **Predictive Modeling:** Integrating Machine Learning to "Predict" which customers might move from "Low Risk" to "High Risk" in the next quarter.
- **Automated Alerting:** Setting up Power BI "Data Alerts" to notify managers via email whenever a regional fraud flag threshold is exceeded.

5.5 Conclusion

This project successfully analyzed the dataset to derive meaningful insights and support data-driven decision-making. Through data cleaning, processing, and visualization, important patterns related to customer behavior and business performance were identified.

The use of Excel and Power BI helped in transforming raw data into interactive dashboards and clear visual reports. Additionally, the integration of AI tools improved efficiency in data analysis, report generation, and insight extraction. The project demonstrates how data analytics techniques can be effectively used to solve real-world business problems.

Overall, this project bridges the gap between raw data and strategic intelligence. By combining Excel's calculation power, Power BI's visualization, and Copilot's conversational ease, we have created a comprehensive tool for modern financial management. The system successfully meets all objectives, proving that visual and AI-driven data exploration is the future of the banking industry.

REFERECES

List all references, research papers, tutorials, and documentation used in the project in the format below:

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